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ZENER DIODE AS VOLTAGE REGULATION

AIM:-

To simulate the line and load regulation operation zener diode.

APPARATUS REQUIRED:-

Laptop with Protous software.

THEORY:-

A heavily dopped PN junction diode is called zener diode. Due to heavily dopped nature zener diode works under forward bias and reverse bias conditions.

zener diode under reverse bias conditions is used as a voltage regulation. Even if the input voltage changes the output is constant. This is called line regulation.

If the load resistance changes the output voltage is constant this is called load regulation.

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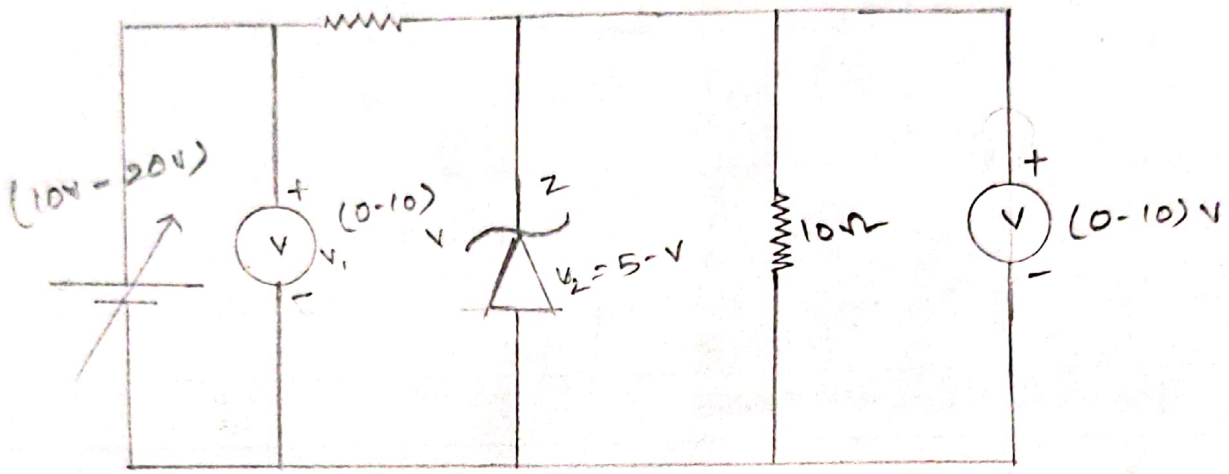
CIRCUIT DIAGRAM:-

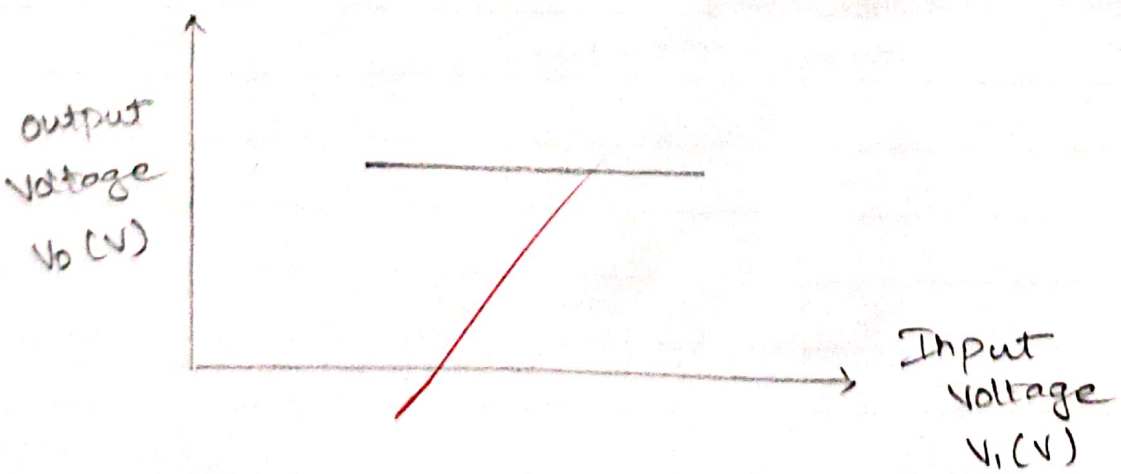
LINE REGULATION



TABULATION:-

S.No	Input Voltage	Output Voltage
1	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.18
6	+17.0	+5.19

MODEL GRAPH:-



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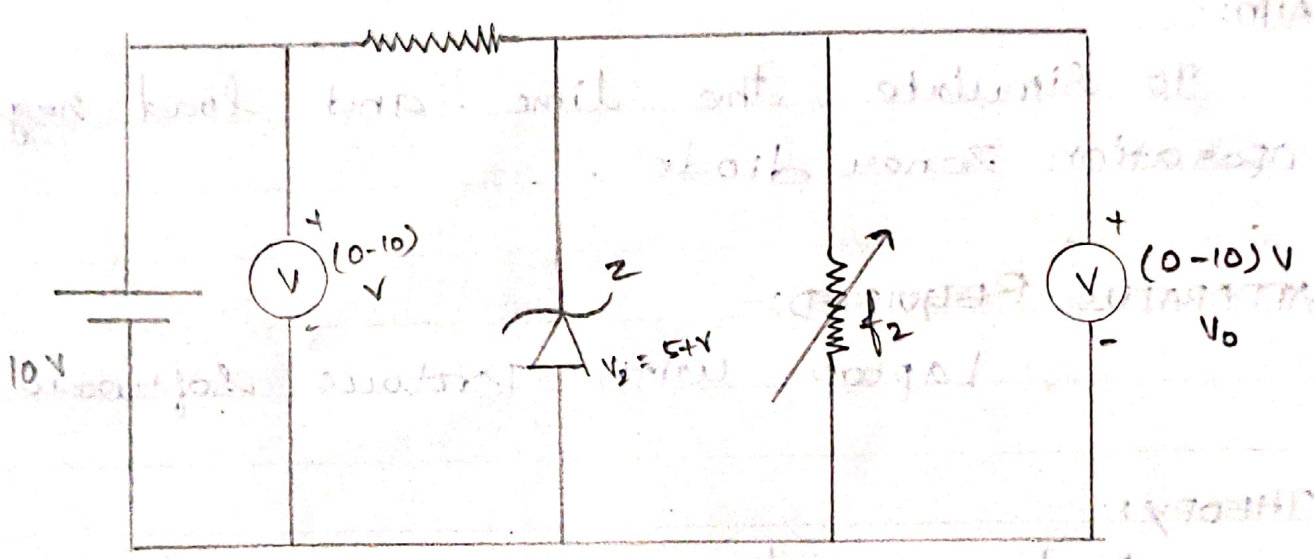
CIRCUIT DIAGRAM:-

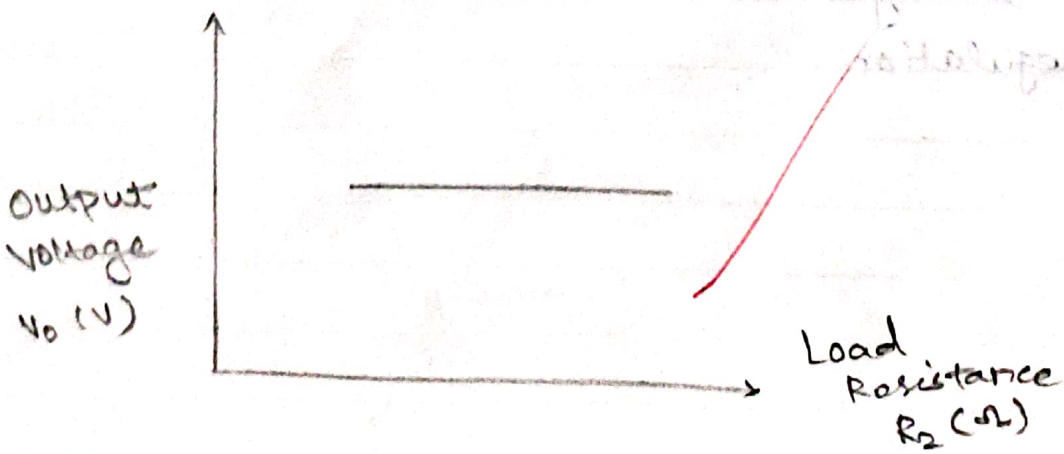
LOAD REGULATION



TABULATION:-

S.No	Load Resistance	Output Voltage
1	100Ω	+5.19
2	200Ω	+5.20
3	300Ω	+5.20
4	400Ω	+5.20
5	500Ω	+5.20
6	600Ω	+5.21

MODEL DIAGRAM:-



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PROCEDURE:-

1. Drag the required components from the proteus library.

2. connect the components (required) as per circuit diagram.

3. For line regulation vary the input voltage and note the output voltages in tabulation. Draw the graph between input voltage and output voltage.

4. For load regulations vary the load resistance and note the output voltage in tabulations. Draw the graph between load resistance and output voltage.

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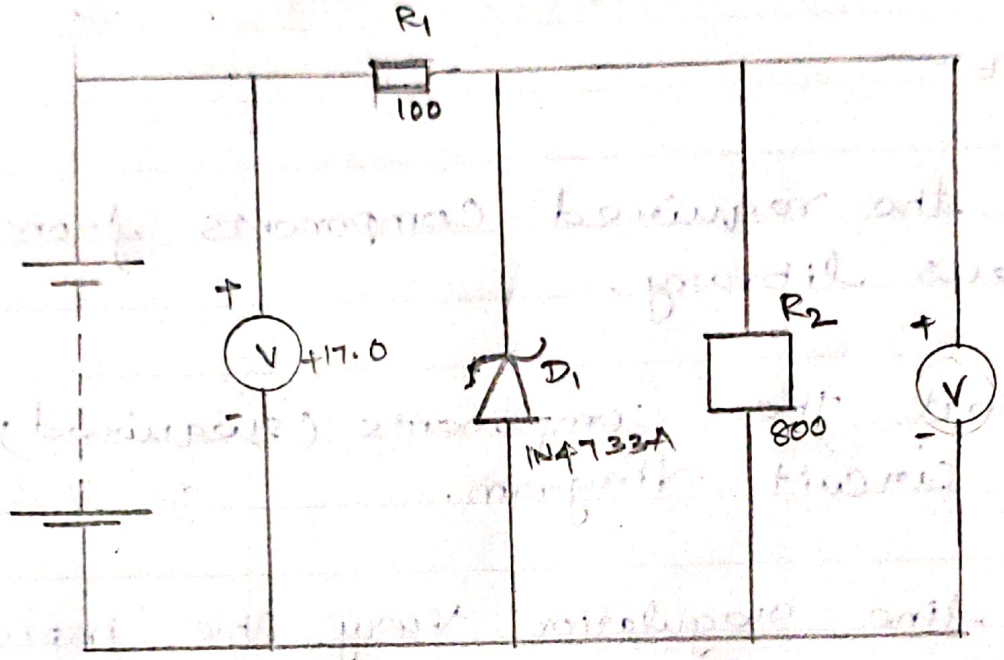
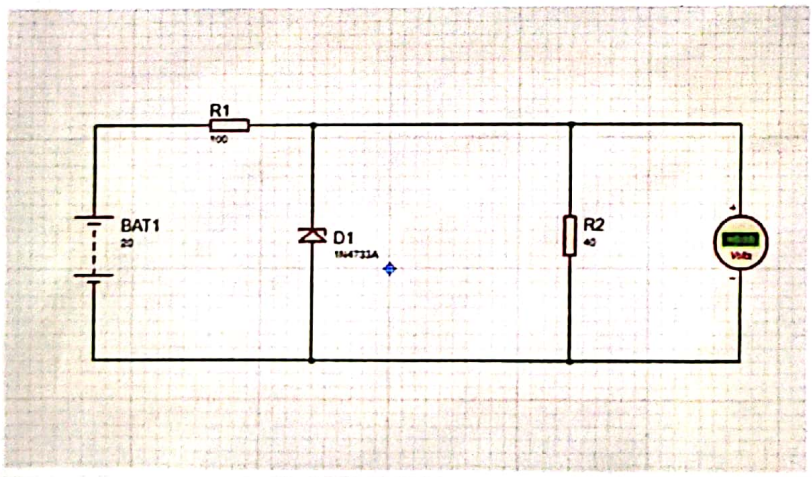
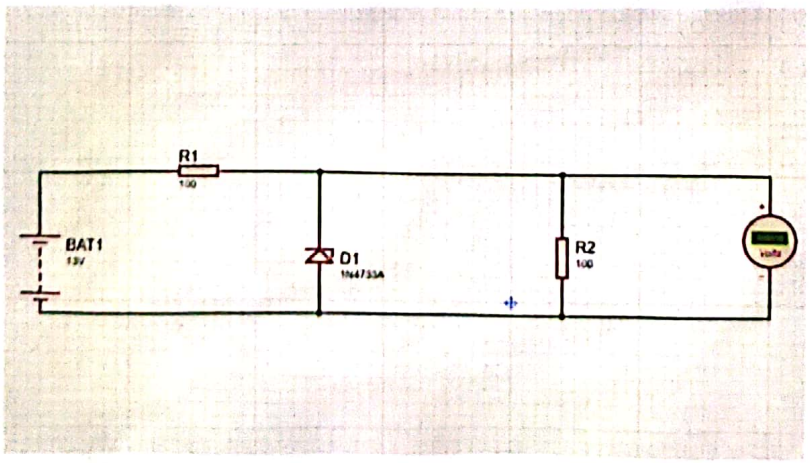
OUTPUT:







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RESULT:-

Thus stimulation of line load regulation operation of zener diode successfully completed.

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